

Zachary West

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EDUCATION

University of Alabama

Tuscaloosa, AL

Bachelor of Science in Computer Engineering – GPA: 3.92/4.00

Aug. 2023 – May 2027

- Minor in Computer Science and Mathematics
- Relevant Coursework: Data Structures and Algorithms, Electronics, Signals and Systems, Embedded Systems
- Activities: Undergraduate Researcher under Dr. Zhengnan Li, President of ABXY Gaming Club

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, HTML5/CSS, Assembly

Developer Tools: Git, Visual Studio Code, MPLAB, STM32CubeIDE, STM32 HAL, Version control

Libraries: NumPy, Pnoise2, Matplotlib, Hashlib

EXPERIENCE

Guitar Pedalboard — UA Undergraduate Research

Aug 2025 – Present

Founder, Project Manager, Design Lead, Software Lead

Tuscaloosa, AL

- Founded, designed, and led a research project creating a pedalboard that models room-specific acoustic effects.
- Researched DSP techniques under Dr. Zhengnan Li, focusing on estimating time-variant impulse responses.
- Developed system architecture, DSP signal flow, and embedded software for real-time audio processing.
- Updated, installed, and maintained over-the-air acoustic communications testbed for remote simulation

HealthTrust, HCA

Jun 2024 – Aug 2024

Executive Intern

Nashville, TN

- Reported to President/CEO, shadowed in executive meetings including MORs, AI workshop, dashboard meetings.
- Shadowed Executive Leadership including CPO, GPO CFO, Director of Information Security as well as the Supply Chain Financial Ops VP in various meetings, projects, and 1:1 mentor meetings.
- Streamlined onboarding for 600+ committee board members by building Excel and Microsoft Forms workflows with version control, dynamic email templates, and cloud-based sharing, **eliminating 50+ hours of manual work.**
- Automated PTO reporting for a team of 20, eliminating manual data entry and email communication.

Gasoline Inc.

Jan 2019 – Current

Apprentice, Event Technician, A1 Grip

Nashville, TN

- Collaborated in professional studio as a **performer and assistant on a Grammy Award-winning album**
- Filled various technical assistant roles in projects such as a documentary, TV Pilot, and video podcast
- Installed equipment on property including a Sauna, Cold Plunge, 25ft flagpole, and an electrical panel

PROJECTS

Money Trees | *Python, Flask, OpenAI API, SQLite, React, Next.js, Chart.js,*

- **1st Place Fintech Winner – UAIInnovate Hackathon:** Developed an AI-powered personal finance dashboard.
- Developed a prompt system (OpenAI API) for personalized financial advice and simplified complex terms.
- Built budget optimization microservices with Node.js and designed banker-friendly UI praised by judges.

Byte History | *Python, OpenAI API, Matplotlib, NumPy, Hashlib, Pnoise2, Voronoi, NLTK*

- Designed and implemented a Python-based procedural world generation engine, producing unique 2D maps using string-based random seeds, Perlin noise, Voronoi seeding, and tectonic plate simulation.
- Generated altitude determined by Perlin Noise with influences from tectonic plate fault lines
- Visualized worlds with Matplotlib, featuring interactive zoom/pan and dynamic coloring
- Enabled deterministic random seeding via string hashing, ensuring reproducible world generation.

Chess Game | *C++ , bitset*

- Engineered a playable chess game in C++ with an emphasis on object-oriented programming and bit manipulation, using bitboards for move generation, board representation, and rule enforcement.
- Implemented a move parser to handle algebraic notation input and generate move objects with correct metadata.